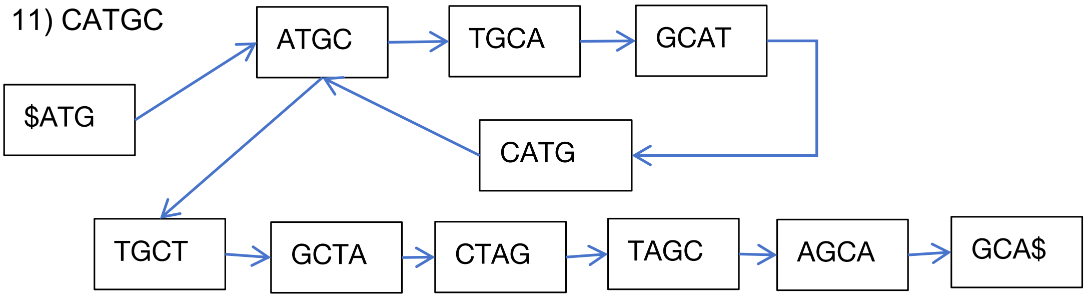
1. Assembly and de Bruijn graph. Given the following reads, draw the de Bruijn graph (4-mer as node), and write down what is the whole sequence.

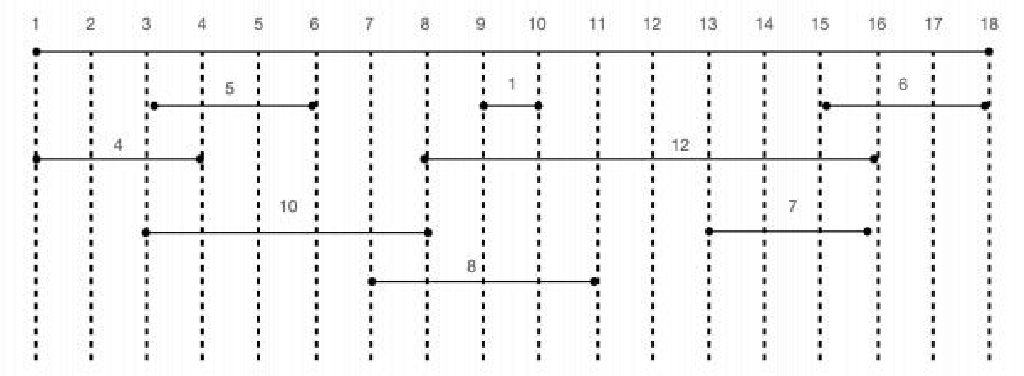
1) $ATGC
2) TGCAT
3) ATGCT
4) GCTAG
5) CTAGC
6) TAGCA
7) AGCA$
8) ATGCA
9) GCATG
10) TGCTA
11) CATGC

ANS:

ATGCATGCTAGCA



2. Here is the position and weight of all exon segments in an abstract sequence of length 18. Please find a maximum set of non-overlapping putative exons based on brute force or dynamic programming.



ANS:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
0	0	0	4	4	5	5	10	10	11	13	13	13	13	13	22	22	22

3. **Please generate a dynamic programming matrix for sequence alignment and find its optimum alignment score.**

Sequence_A: GATTCATA Sequence_B: GGATCGA

Scoring system: +5 for a match; -2 for a mismatch; -6 fore each indel

ANS:

		G	G	A	T	C	G	A
	0	-6	-12	-18	-24	-30	-36	-42
G	-6	5	-1	-7	-13	-19	-25	-31
A	-12	-1	3	4	-2	-8	-14	-20
T	-18	-7	-3	1	9	3	-3	-9
T	-24	-13	-9	-5	6	7	1	-5
C	-30	-19	-15	-11	0	11	5	-1
A	-36	-25	-21	-10	-6	5	9	10
T	-42	-31	-27	-16	-5	-1	3	7
A	-48	-37	-33	-22	-11	-7	-3	8

p.s.Each cell depends only on the values of its three adjacent cells: top-left, top, and left.

For $i \geq 1, j \geq 1$:

$$dp[i][j] = \max \begin{cases} dp[i-1][j-1] + \begin{cases} +5, & \text{if } A[i] = B[j] \\ -2, & \text{if } A[i] \neq B[j] \end{cases} \\ dp[i-1][j] - 6 & \text{(A[i] corresponds to a gap)} \\ dp[i][j-1] - 6 & \text{(B[j] corresponds to a gap)} \end{cases}$$

4. Homology Search and Phylogenetic Analysis of MLH1

Objective:
Use BLAST, multiple sequence alignment, and phylogenetic tree construction to explore the evolutionary relationship of the **human MLH1** gene across species.
Tasks:
1. Use **BLASTp** to search homologous protein sequences of MLH1 from the NCBI.

2.Select representative homologs from diverse species (about10–15 sequences).
3.Download the protein sequences in FASTA format and combine them into onefile.
4.Perform **multiple sequence alignment** (MSA) using your preferred alignment tool
5.Construct a **phylogenetic tree** using your aligned sequences.
6.Visualize and annotate the tree.
Submit:

•The alignment file(`.fasta`),
•The phylogenetic tree (image), and
•A short summary (within one page) describing your findingsand observations.

ANS:

Guide Tree

```
(
(
XP_069173196.1:0.19427
,
XP_069173195.1:0.19427
):0.057947
,
(
XP_069173199.1:0.263911
,
(
(
XP_069173193.1:0.234994
,
(
XP_069173202.1:0.158525
,
(
XP_069173200.1:0.229124
,
(
XP_069173197.1:0.205825
,
(
XP_069173192.1:0.101404
,
(
XP_069173194.1:0.0686167
,
XP_069173204.1:0.0686167
):0.0327877
):0.10442
):0.0232996
):-0.0705992
):0.0764685
):0.00645865
,
(
XP_069502281.1:0.212858
,
(
XP_069493420.1:0.184293
,
CAI5525238.1:0.184293
):0.0285652
):0.028594
):0.0224584
):-0.0116937
)
;
```

Phylogram